

Scaling Laws for In-Context Bayesian Near-Field Linear Sampling

Technical results note

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Abstract

We study how an in-context posterior-moment solver for the original two-dimensional near-field linear sampling method scales with the number n of training inverse problems, network size P , context length m , recurrent depth, and wall-clock time. The input is a complex multistatic response from deterministic point sources around one to six sound-soft obstacles. A fixed von Mises/softmax covariance selects the Gaussian-process feature space; its nonlinearity is a modelling choice and is not learned. Richardson, heavy-ball (HB), Chebyshev and PCG share the same context encoder and Bayesian decoder and solve posterior mean and covariance right-hand sides in parallel. Identity- preconditioned CG is the matched-matrix-vector-product control. The sweep contains 2 trained checkpoints and 336 learned- model/task evaluations. We report task-level uncertainty, inference time, empirical scaling exponents and a simultaneous distribution-free held-out risk bound. The main question is not whether a large controller can imitate CG, but when learned conditioning improves CG under severe, task-dependent ill-conditioning.

1 Problem and posterior moments

Let $N_D \in \mathbb{C}^{m \times m}$ be the whitened near-field matrix for a sound-soft obstacle D , $\Phi = [\phi_z]_z$ the Green-function probes on a two-dimensional sampling grid, and $K_\Gamma \succ 0$ a prescribed angular GP covariance. The nonlinear feature choice is

$$W_{ij} = \exp\{\gamma \cos(\theta_i - \theta_j)\}, \quad K_\Gamma = (1 - \eta)I + \eta D_W^{-1/2} W D_W^{-1/2},$$

with fixed $\gamma = 1$ and $\eta = 0.2$. Define

$$H_D = N_D K_\Gamma N_D^* + I, \quad H_D Q_\mu = \Phi, \quad H_D Q_\Sigma = N_D K_\Gamma.$$

One tied recurrence acts on $[\Phi, N_D K_\Gamma]$ and yields

$$M_D = K_\Gamma N_D^* Q_\mu, \quad \Sigma_D = K_\Gamma - K_\Gamma N_D^* Q_\Sigma.$$

Thus a small mean residual is insufficient for Bayesian correctness: the covariance residual must converge as well.

2 Variable-context architecture and matched solvers

The covariance block uses a deterministic orthonormal harmonic sketch with a fixed number of columns and m rows. Its feature dimension is therefore independent of context length, while the physical operator is never padded or replaced by a learned surrogate. Training contexts are [12, 16, 24, 32]; evaluation contexts are [8, 12, 16, 24, 32, 48], so the endpoints include context-length extrapolation. Controller widths are [128] and sample sizes are [128, 256].

At test depth $T = 32$, all recurrent methods use exactly T applications of the transformed operator for both posterior blocks. Identity-CG sets the population factor to I . Population-PCG learns a positive geometry/population factor and retains exact task-wise Krylov coefficients. Looped HB, Chebyshev and Richardson additionally infer finite-horizon spectral endpoints from RHS-weighted Krylov statistics. Consequently, PCG versus CG isolates learned conditioning; a looped method versus PCG measures whether a stationary learned recurrence is worth replacing Krylov adaptation.

3 Axes and scaling model

The dataset size n is the number of unique physical inverse problems presented once in an online stream; the eight examples in an optimizer batch share one sampled acquisition configuration. The context length m is both the

Table 1: Empirical exponents in Eq. (1) for both risks; larger positive values indicate faster improvement along that axis.

Method	risk	data α	context β	width γ	R^2
population-PCG	localization	0.010	3.000	2.194	0.984
population-PCG	solver	0.010	0.034	0.213	-0.096

Table 2: Final comparison at context $m = 24$, width 128 and $n = 256$. Baselines do not train. Runtime is milliseconds per held-out batch.

Method	AP	mean res.	covariance res.	score error	ms
population-PCG	0.768	0.213	6.25e-03	3.97e-03	21.291
identity-CG	0.769	0.213	6.50e-03	4.11e-03	15.106
exact	0.768	5.98e-05	1.65e-06	0	10.968

number of point sources and receivers, giving m^2 complex measurements. Network size P counts trainable parameters. Each held-out configuration contains fixed, never-trained MFS tasks spanning ID obstacles and OOD obstacle count, shape, noise, aperture and frequency.

For localization risk $R_{\text{loc}} = 1 - \text{AP}$ and bounded numerical risk $R_{\text{solve}} = r/(1+r)$, we separately fit the descriptive law

$$R(n, m, P) = R_\infty + A(n/n_0)^{-\alpha} + B(m/m_0)^{-\beta} + C(P/P_0)^{-\gamma} + D\sqrt{(P/P_0)/(n/n_0)}. \quad (1)$$

The last term allows a finite-data width penalty. Equation (1) is an empirical scaling model, not a theorem.

4 A simultaneous held-out generalization bound

Let $L_c \in [0, 1]$ be localization risk for a pre-specified trained configuration $c = (n, m, P, \text{method})$, and let $\hat{R}_c$ average T_c independent held-out tasks. A union bound and one-sided Hoeffding inequality give the following non-asymptotic certificate.

Theorem 1 (Simultaneous task-distribution bound). *For K evaluated configurations, with probability at least $1 - \delta$, simultaneously for every c ,*

$$R_c \leq \hat{R}_c + \sqrt{\frac{\log(K/\delta)}{2T_c}}.$$

We use $\delta = 0.05$ and report the clipped upper bound in Figure 1. Training samples determine the learned hypothesis and hence $\hat{R}_c$; held-out sample size controls certification slack. The same theorem is applied to the bounded numerical risk $L_c^{\text{solve}} = r_c/(1+r_c)$, where r_c is the posterior-mean relative residual. This second risk remains sensitive after localization AP saturates. This distinction matters. Replacing T_c by training size n after fitting would be invalid, while a generic parameter-count uniform bound for this nonconvex complex network is vacuous at the present P/n ratios.

5 Results

The equal-budget residual ratios at the final central configuration are $1.004\times$ for the posterior mean and $1.039\times$ for posterior covariance. Ratios above one isolate a benefit from the learned population factor beyond ordinary CG. Localization can nevertheless saturate before residual convergence; in that regime additional solver accuracy does not overcome the physical point-spread limit.

6 Interpretation and limitations

Three claims are deliberately separated. First, CG is a strong task-adaptive baseline because its coefficients already depend on the current Krylov space. Second, population-PCG tests whether geometry-distribution conditioning adds value when no obstacle-specific preconditioner is known. Third, looped HB tests amortized fixed-depth iteration, not a universally superior replacement for CG. A true task-specific ICL preconditioner would additionally map current

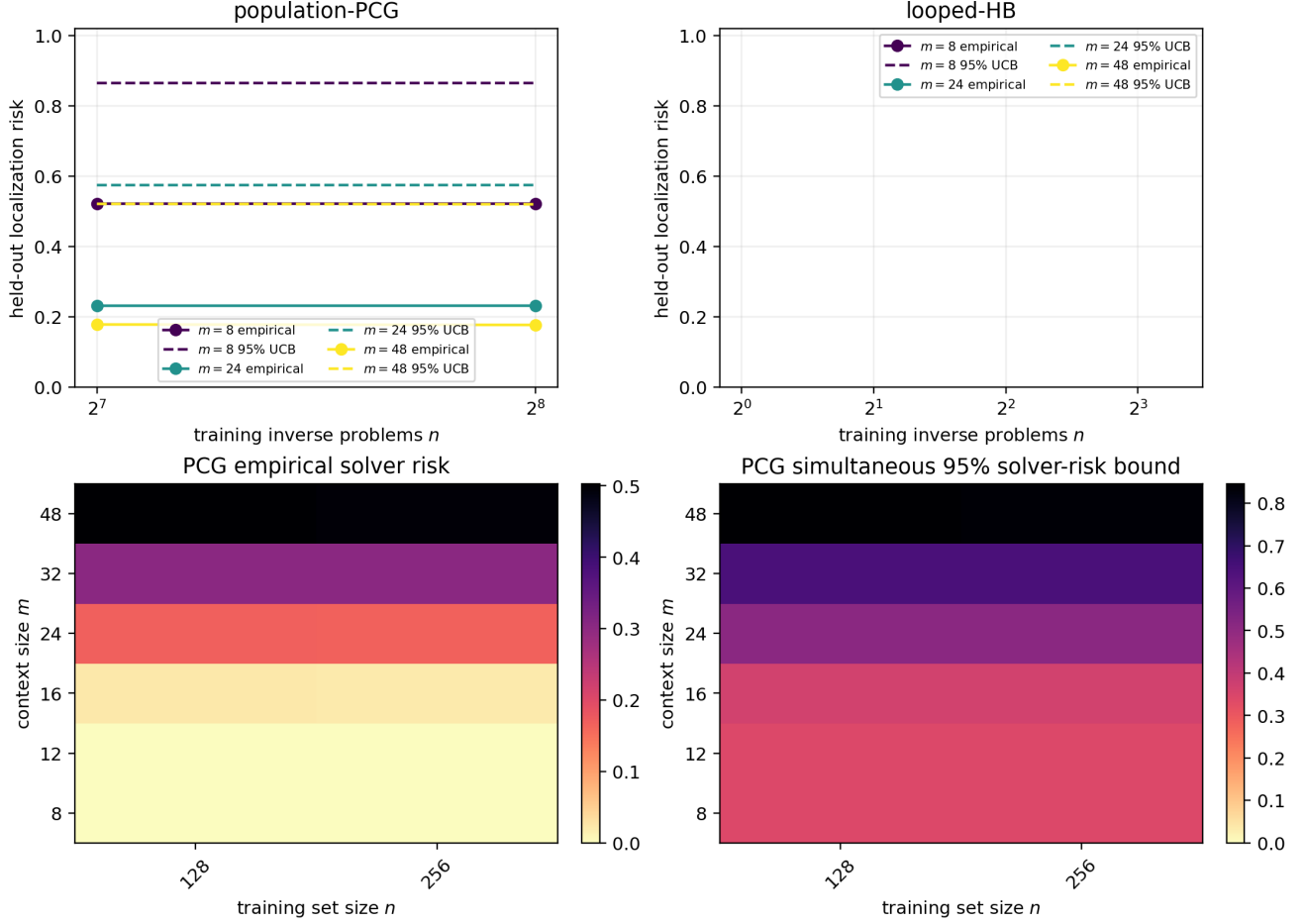


Figure 1: Empirical localization and numerical solver risks with simultaneous 95% held-out upper bounds across dataset and context sizes.

moment or Ritz information into an SPD operator $C_D(A_D) \approx A_D^{-1/2}$; the present population factor does not rotate all obstacle-dependent eigendirections.

The held-out bound certifies the stated simulator distribution only. The forward data are generated by MFS, not laboratory measurements, and context extrapolation does not imply robustness to arbitrary receiver geometries. Physical resolution remains controlled by aperture, frequency and sensor count. These qualifications prevent a small numerical loss from being misreported as solved inverse scattering.

7 Reproducibility

All task-level records, cumulative training times, checkpoints, runtime measurements, scenario definitions, fitted parameters and bound values are stored beside this PDF. The experiment uses three fixed seeds and preserves the exact same held-out tasks across methods, widths and dataset checkpoints.

References

- [1] G. Kang, C. J. Lee, and X. Cheng. Transformers can learn posterior predictive distributions in-context. *arXiv:2605.26713*, 2026.
- [2] J. Garnier, H. Haddar, and H. Montanelli. The linear sampling method for random sources. *arXiv:2210.15560*, 2022.

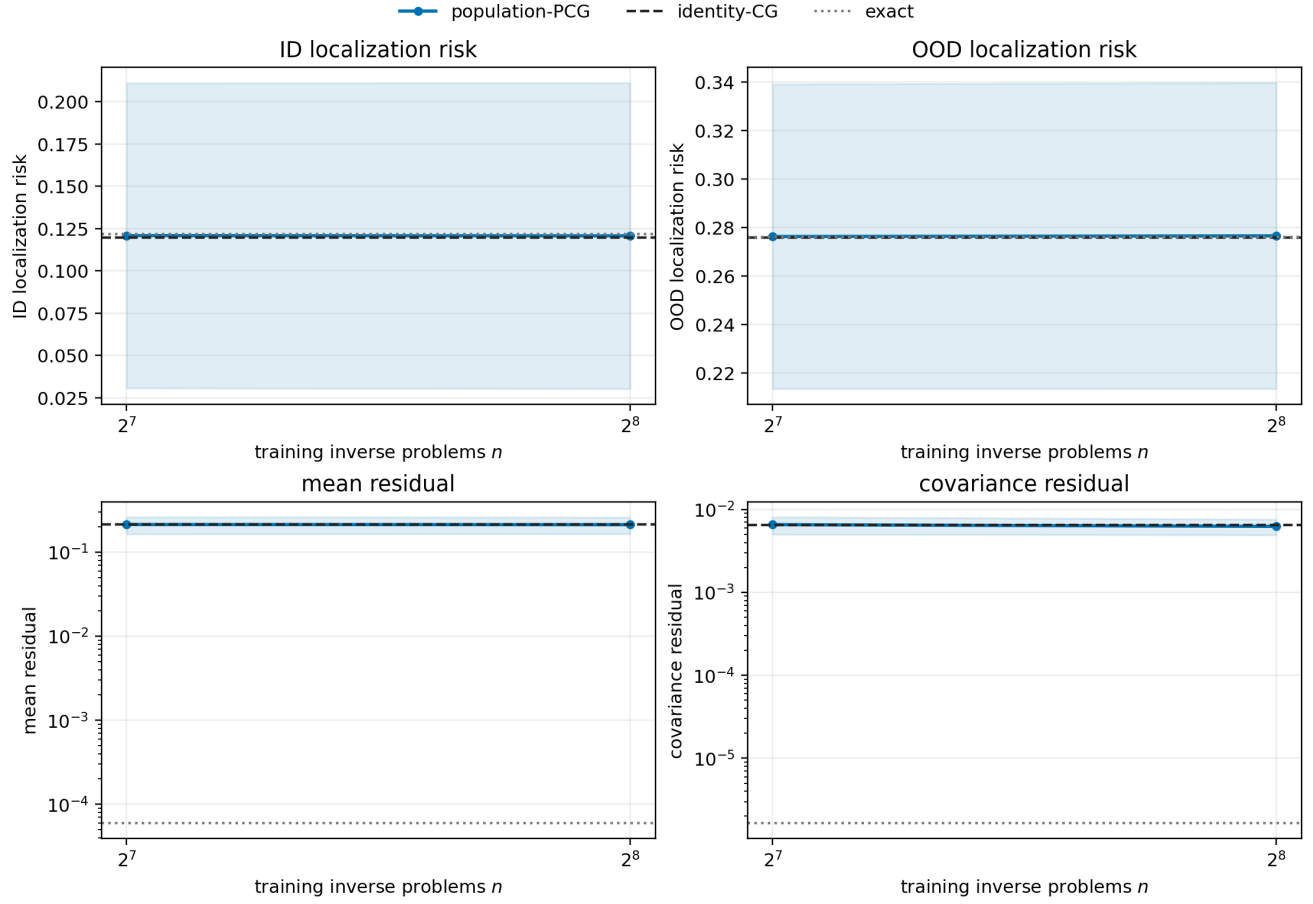


Figure 2: Learning curves in dataset size at fixed network width and context.

- [3] J. Garnier, H. Haddar, and H. Montanelli. The linear sampling method for data generated by small random scatterers. *arXiv:2403.19482*, 2024.
- [4] B. T. Polyak. Some methods of speeding up the convergence of iteration methods. *USSR Computational Mathematics and Mathematical Physics*, 1964.

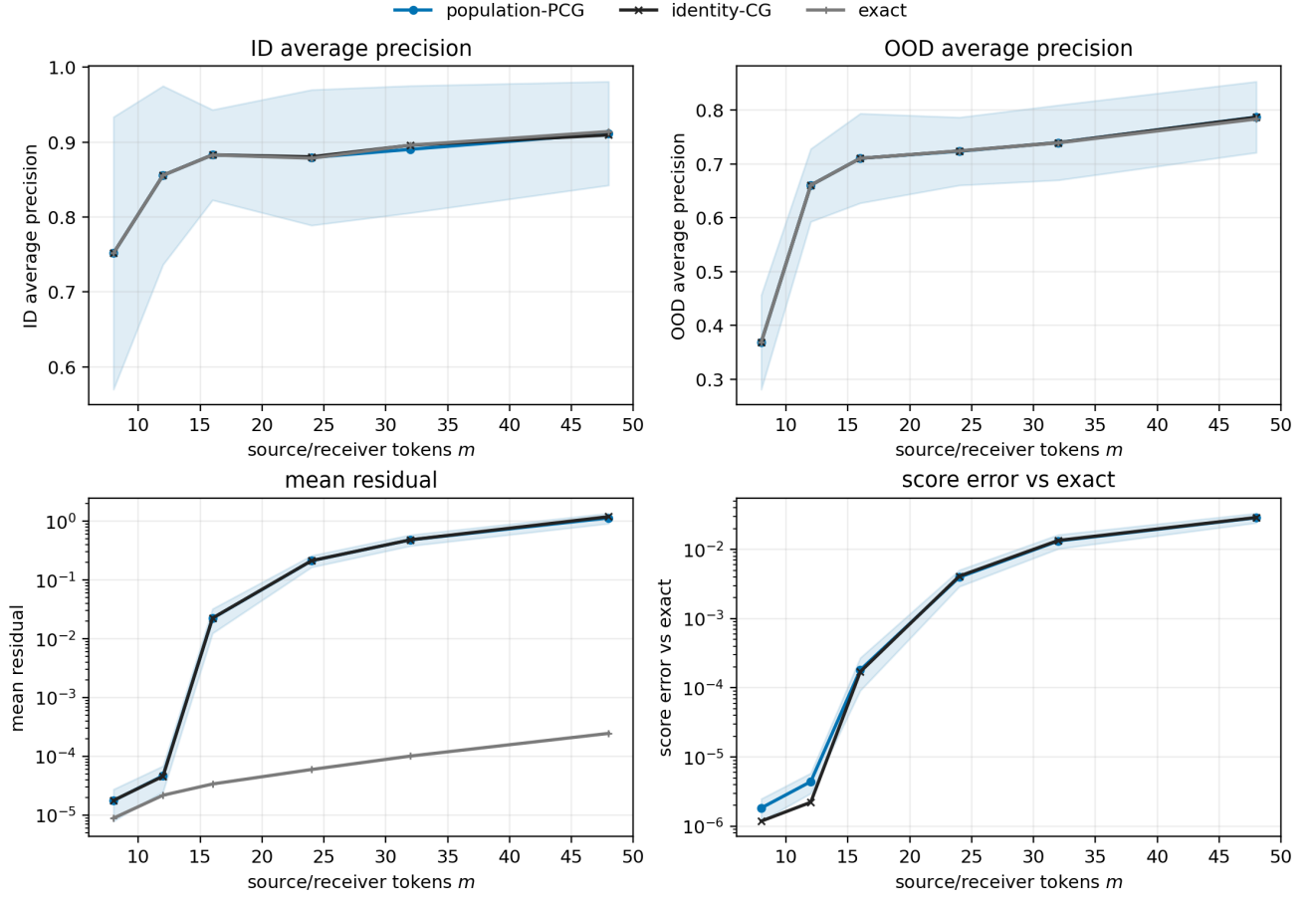


Figure 3: Context-length scaling. Context $m = 8$ and $m = 48$ are outside the training context set.

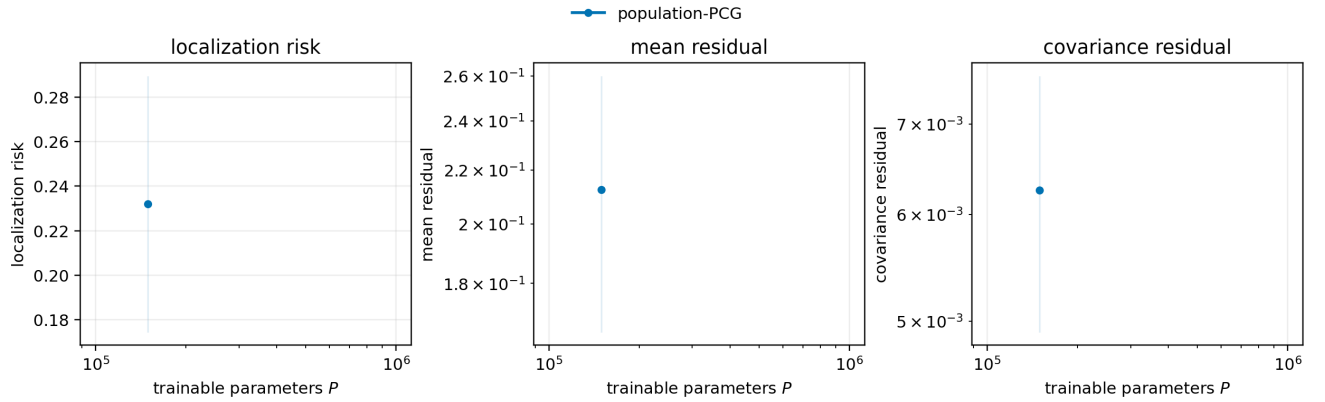


Figure 4: Network-size scaling at the largest completed dataset size.

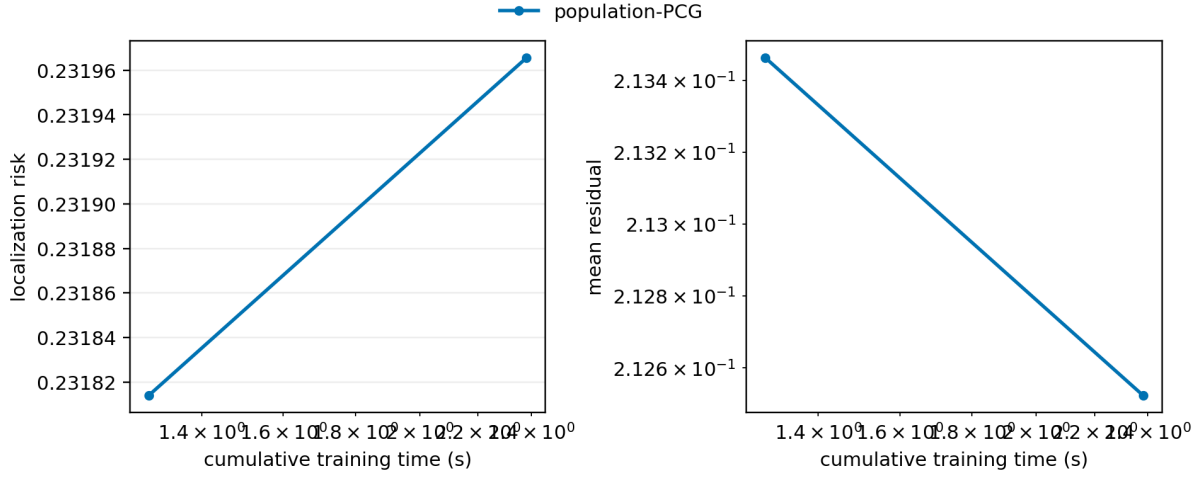


Figure 5: Wall-clock learning curves. Time includes physical task generation, forward/backward propagation and optimization, but excludes held-out audits.

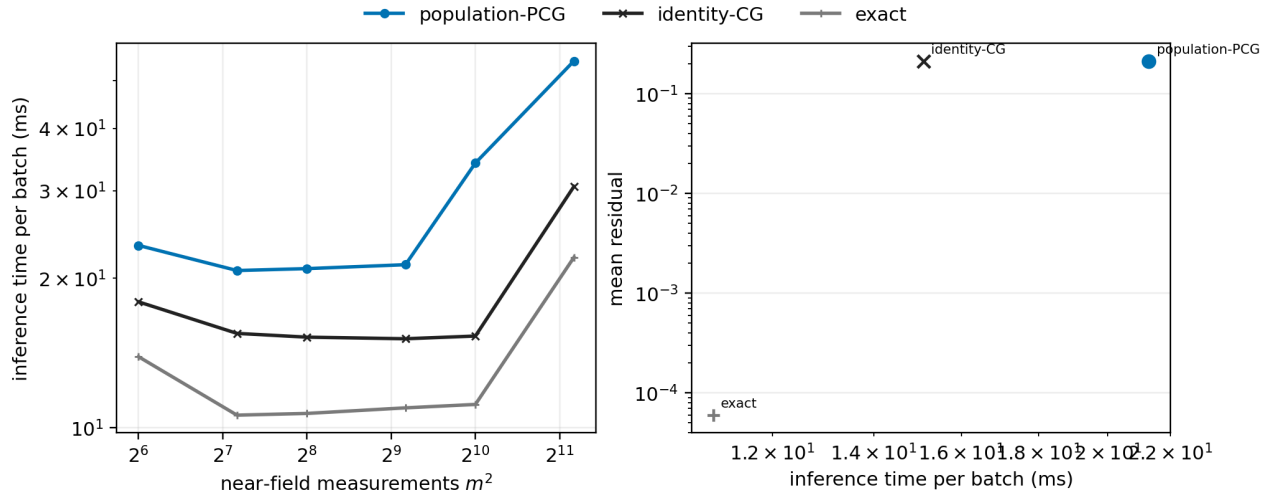


Figure 6: Inference scaling with m^2 complex measurements and the residual–time Pareto comparison at $m = 24$.